

Experiment Report: Exp 2-8 (COMPLETE RESULTS)

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ALL experiments use SAME test set (481 images)

Complete Experiment Results (Same Test Set):

Exp	Model	Test Acc	P(Orig)	R(Orig)	Status
2	Fourier_Balanced_CNN	0.499	0.000	0.000	FAILED
3	Fourier_MobileNetV2	0.499	0.000	0.000	FAILED
4	Laplacian_MobileNetV2	0.846	0.835	0.863	GOOD
5	Hybrid_MobileNetV2	0.807	0.968	0.635	IMBALANCED
6	RGB_MobileNetV2	0.857	0.814	0.925	GOOD
7	RGB_EfficientNetB0	0.877	0.870	0.888	BETTER
8	RGB_Laplacian_EfficientNetB0	0.900	0.910	0.880	BEST

Note: Exp 5 (Hybrid) uses Fourier+Laplacian preprocessing
P(Orig) = Precision for Originals, R(Orig) = Recall for Originals
Test set: 481 images (241 originals, 240 recaptures)
FAILED = Model always predicts one class (class collapse)

Detailed Class Performance:

Exp	Model	Acc	P(O)	R(O)	P(R)	R(R)
2	Fourier_Balanced_CNN	0.499	0.000	0.000	0.499	1.000
3	Fourier_MobileNetV2	0.499	0.000	0.000	0.499	1.000
4	Laplacian_MobileNetV2	0.846	0.835	0.863	0.858	0.829
5	Hybrid_MobileNetV2	0.807	0.968	0.635	0.728	0.979
6	RGB_MobileNetV2	0.857	0.814	0.925	0.913	0.788
7	RGB_EfficientNetB0	0.877	0.870	0.888	0.885	0.867
8	RGB_Laplacian_EfficientNetB0	0.900	0.910	0.880	0.880	0.920

P(O)=Precision Originals, R(O)=Recall Originals, P(R)=Precision Recaptured, R(R)=Recall Recaptured

Exp 2: Fourier_Balanced_CNN	-> 49.9% (COMPLETE FAILURE: Always predicts recaptured)
Exp 3: Fourier_MobileNetV2	-> 49.9% (COMPLETE FAILURE: Same pattern as Exp 2)
Exp 4: Laplacian_MobileNetV2	-> 84.6% (FIRST WORKING: Balanced P/R both classes)
Exp 5: Hybrid_MobileNetV2	-> 80.7% (IMBALANCED: High P/low R originals)
Exp 6: RGB_MobileNetV2	-> 85.7% (IMPROVED: Better than feature engineering)
Exp 7: RGB_EfficientNetB0	-> 87.7% (BETTER ARCH: More balanced)
Exp 8: RGB_Laplacian_EfficientNetB0	-> 90.0% (BEST: Perfect F1 balance)

Critical Technical Findings:

1. FOURIER PREPROCESSING DESTROYS MODEL LEARNING:

- Experiments 2 & 3: Precision_originals = 0.0, Recall_originals = 0.0
- Fourier transform eliminates recapture artifacts needed for classification
- Models collapse to single-class prediction (always 'recaptured')

2. LAPLACIAN (EDGE DETECTION) PRESERVES IMPORTANT FEATURES:

- Experiment 4: 84.6% accuracy with balanced performance
- Edge features capture recapture boundary artifacts
- But: RGB images work even better (85.7% vs 84.6%)

3. HYBRID FOURIER+LAPLACIAN CAUSES IMBALANCE:

- Experiment 5: 80.7% accuracy but severe class imbalance
- Originals: P=0.968 (high), R=0.635 (low) -> many false negatives
- Recaptured: P=0.728 (low), R=0.979 (high) -> many false positives

4. RGB-ONLY OUTPERFORMS FEATURE ENGINEERING:

- Exp 6 (RGB MobileNetV2): 85.7% > Exp 4 (Laplacian): 84.6%
- Modern CNNs learn optimal features directly from raw images
- Feature engineering may remove important information

5. EFFICIENTNETB0 ARCHITECTURE SUPERIORITY:

- Exp 7 (EfficientNetB0): 87.7% > Exp 6 (MobileNetV2): 85.7%
- Compound scaling and better design yield +2% improvement
- More consistent CV performance (lower variance)

6. OPTIMAL COMBINATION: RGB + LAPLACIAN + EFFICIENTNETB0:

- Experiment 8: 90.0% accuracy (best overall)
- Perfect F1 score balance (0.90 for both classes)
- Combines raw image information with edge-enhanced features

Cross-Validation Insights:

1. Fourier Models (Exp 2-3):

- Inconsistent class prediction across folds
- Complete failure on test set

2. Laplacian MobileNetV2 (Exp 4):

- High variance: 55.7% to 81.5% across folds
- Fold 3 shows severe recall imbalance

3. Hybrid Model (Exp 5):

- Highest variance: 65.6% to 84.6% across folds
- Most unstable model architecture

4. RGB Models (Exp 6-8):

- Lower variance and more consistent performance
- EfficientNetB0 shows best CV consistency